

AUTHENTIC LANKA EXPORTS ERP

Comprehensive Customer User Manual

Welcome to the Authentic Lanka Exports ERP (Manufacturing & Distribution) system. This user manual provides a comprehensive, end-to-end guide to navigating and operating the ERP console effectively.

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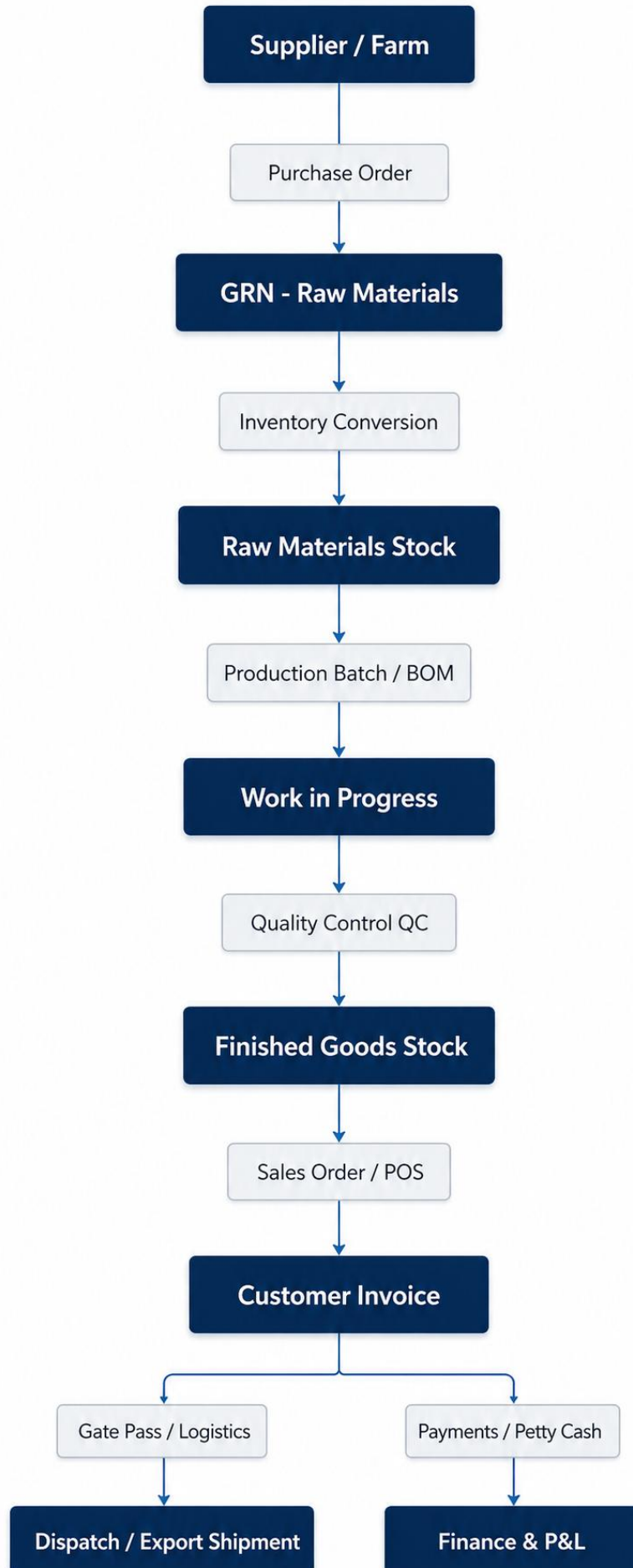
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1. System Overview & Architecture

The Authentic Lanka Exports ERP is an integrated environment designed to manage corporate manufacturing, warehouse inventory pipelines, sales, logistics operations, and human resources inside a single centralized cross-functional database hub.

Data Flow Overview

Data flows seamlessly across operational modules to eliminate traditional human indexing errors and cross-department balance discrepancies:



Technology Highlights

- **Real-Time Data:** Driven by a state-of-the-art React-Vite frontend workspace and an Express-Node architecture backend layer, utilizing Socket.io connection tunnels to push live real-time updates (e.g., immediate stock reductions and active automated notifications) across all open user terminals.
- **Audit Trail:** Every record modification is written to the system audit log database table, capturing the employee ID, exact server timestamp, target module, action type, and historical before/after values.

2. Login, Credentials & Security Controls

Every employee must authenticate securely using their uniquely assigned enterprise structural credentials.

Default Developer / Testing Credentials

For deployment verification and staging runs, the database comes pre-seeded with a default Administrator credential rule:

- **Email Address:** `admin@admin.com`
- **Default Password:** `Admin123!`
- **Access Level:** Full Access (Admin Configuration Module)

Login Security Policies

- **Password Visibility:** Click the Eye icon on the right side of the password field to toggle character visibility and prevent entry errors.
- **Failed Login Attempts Counter:** You are allowed up to 5 failed login entries. If an validation fails, the interface triggers an alert stating: `"Invalid email or password. X attempt(s) remaining."`
- **Account Lockout:** After 5 consecutive failed verification attempts, the account profile is locked for **15 minutes** to prevent brute-force perimeter penetration. The system displays: `"Account locked due to too many failed attempts. Try again in 15 minutes."`
- **Deactivated Users:** Inactive or suspended accounts are restricted at routing level. The user is greeted with: `"Account is deactivated. Contact admin."`

3. User Roles & Access Permissions Matrix

The system relies heavily on strict **Role-Based Access Control (RBAC)** matrices to enforce operational security limits across business domains.

Core Roles Overview

- **Admin / Super Admin:** Complete access, user account registration, security role configurations, database backups, global settings, and audit log tracking charts.

- **Warehouse Manager:** Master catalog access, product category/brand mapping, multi-warehouse location allocations, stock transfers, inventory adjustments, and vendor supplier returns.
- **Warehouse Staff:** View stock quantities, record incoming Goods Receipt Notes (GRNs), perform internal transfers, and log stock damage sheets.
- **Production Staff:** Manage recipes (Bills of Materials), log manufacturing runs (batches), track raw ingredient consumption weight, monitor utility drawings, and record final factory batch yields.
- **Sales Manager:** Review and approve commercial sales orders, apply credit-limit bypass overrides, configure POS settings, issue tax invoices, process product returns, and generate analytics sheets.
- **Sales Rep:** Create downstream customer sales orders, log transactions on POS screens, check catalog pricing data, and register incoming customer payments.
- **Accountant:** Process incoming customer invoices, track supplier billing balances, manage financial ledgers, balance bank accounts, process cheque clearances, and monitor the live Daily P&L report.
- **Cashier:** Access POS terminal checkout environments, process retail customer payments, and generate invoices.
- **HR Manager:** Control corporate employee directories, job designations, shift templates, roster scheduling, attendance synchronization, holiday calendars, payroll calculations, and payslip execution workflows.
- **Employee:** General user account for self-service profiles, logging attendance card cycles, submitting formal leave requests, and downloading personal salary payslips.

4. Product Master (Catalog Management)

The Product Master contains all trackable items flowing throughout your physical supply chain. Registered inventory items must be mapped to one of the following four primary categories:

- **Raw Materials:** Unprocessed items (e.g., green tea leaves, raw coconut husks, spices).
- **Semi-Finished Goods:** Dehydrated, milled, or processed intermediate stocks awaiting packaging steps.
- **Finished Goods:** Trade-ready, commercial products ready for client export and domestic dispatches.
- **Packaging Materials:** Structural boxes, jars, bottles, caps, corporate labels, security seals, and wraps.

Product Code Generation Schema

To preserve catalog string integrity, the system auto-generates a unique SKU identity code utilizing a specialized Julian Date schema calculation format:

Format: P-[CategoryShortCode] - [YearShort] [JulianDayOfYear] - [SequenceNo]

Schema Token Definitions:

- **CategoryShortCode:** 3-letter capital classification code (e.g., Raw Materials = `RAW` , Finished Goods = `FNG` , Packaging Materials = `PKG`).
- **YearShort:** 2-digit representation of the active calendar year (e.g., year 2026 logs as `26`).
- **JulianDayOfYear:** 3-digit technical calendar day value of the year (ranging from 001 to 365/366). For example, June 4th computes as `155` .
- **SequenceNo:** A 2-digit increment counter tracking from 01 that automatically resets to 01 daily per item category line.

Example Output Identity: `P-RAW-26155-03` represents the 3rd distinct raw material profile registered within the ERP platform on June 4th, 2026.

5. Procurement & Supply (GRN & Purchases)

This business module administers farm sourcing agreements, vendor relations, and inbound inventory delivery tracking metrics.

Step-by-Step Purchasing Workflow:

1. **Create Purchase Order (PO):** Go to `Procurement → Purchase Orders → Create PO` . Designate a Vendor Supplier, configure delivery timelines, select destination warehouse bins, input catalog items, map unit costs, and click Submit.
2. **Goods Receipt Note (GRN):** When physical shipping haulage arrives at the gate, navigate to `Procurement → Goods Receipt Notes → Receive Goods` .
3. **Inspection & Rejection Logic:** The checking inspector enters total quantities delivered and flags defective materials. The database calculates accepted units using the following algorithmic script:

$$\text{Accepted Quantity} = \text{Received Quantity} - \text{Rejected Quantity}$$

4. **Yield Forecasting:** If a conversion yield template rule exists for the specific item profile, the live GRN display prints a *Live Yield Forecast* calculating expected post-production batch weights.
5. **Perishables & Expiry Tracking:** Record the corresponding Supplier Invoice #, Supplier Delivery Note #, transport logistics vehicle details, driver identification, and batch expiration markers. Click **Confirm Receipt** to add stock to the warehouse bins.
6. **Supplier Bills:** Formally confirmed GRN records automatically compile draft *Supplier Bills* inside the Finance workspace ledger for Accountant auditing and payment release profiling.

6. Manufacturing & MES (Production, BOM & Forecasting)

This environment runs your factory floor operations, tracking processing ingredient conversion ratios, physical workforce counts, energy drawing coefficients, and quality assurance releases.

A. Bill of Materials (BOM) Costing Formula

Prior to executing a run, a recipe matrix (BOM) must be mapped out. The system aggregates real-time unit costs by analyzing processing wastage parameters, labor inputs, and asset depreciation constants:

$$\text{BOM Cost} = \Sigma [\text{Qty} \times (1 + \text{Wastage\%} / 100) \times \text{Unit Cost}] + \text{Labor Cost} + \text{Overhead Cost}$$

- **Wastage %:** Expected mechanical processing shrinkages, material scrap ratios, or thermal moisture reduction loss factors.
- **Labor Cost:** Total logged operator shift hours multiplied by their contract hourly baseline pay rates.
- **Overhead Cost:** Active machine depreciation indices and utility constant overhead configurations.

B. Logging a Production Batch

Navigate to `Production → Production Batches → Log New Batch`.

1. Select target product code profile, raw material origin tracking lines (for strict gate-to-gate traceability), and choose the specific manufacturing step route (e.g., *Washing → Dehydration → Milling → Sifting → Packing*).
2. **Workforce Allocation:** Log total manufacturing operators segmented by active *Day Shift* and *Night Shift* teams.
3. **Resource Usage Logs:** Input active raw input weight metrics (Kg), total dehydrator runtime hours, volume of Firewood Consumed (Kg), and total Electricity Units drawn (kWh).
4. **Output Registration:** Record the finalized post-packaging manufactured weight metric (Kg).

Upon submitting the yield metrics, the core computation script computes the batch performance index value:

$$\text{Batch Efficiency \%} = (\text{Total Output Weight} / \text{Total Input Weight}) \times 100$$

C. Quality Control (QC) Clearance

When marked as structurally completed, the batch state automatically marks as `Pending QC`. It enters the QC checking pipeline, requiring a certified Laboratory Quality Officer to manually submit key parameter values (e.g., moisture metrics %, purity ratios %, microbial analysis scores) and click **Approve Batch**. Only cleared batches are transferred into trading inventory.

D. Production Forecasting Engine

The analytics platform provides historical predictive capabilities for scheduling managers:

- **Linear Regression Modeling:** Computes past manufacturing run indexes (x) against historical yield efficiency percentages (y). The classic least-squares formula calculation script `y = mx + c` forecasts (subsequent batch yield percentages).

- **Moving Average Forecasting:** Computes smooth trend expectations by analyzing performance coefficients over the previous 5 finalized factory production batch lines.
- **Resource Drawing Rates:** Projects upcoming factory fuel configurations utilizing derived consumption coefficients:

$$\text{Firewood Rate} = \text{Firewood Consumed (Kg)} / \text{Total Input Weight (Kg)}$$

$$\text{Electricity Rate} = \text{Electricity Consumed (kWh)} / \text{Total Input Weight (Kg)}$$

7. Sales, CRM, POS & Customer Returns (RMA)

Manages downstream customer acquisition pipelines, invoicing cycles, retail POS terminals, and returns accounting.

A. Commercial Sales Pipeline Steps

Sales Inquiry → Quotation → Sales Order (SO) → Invoice → Dispatch Verification

- **Sales Orders & Credit Blocks:** If a sales transaction forces a client profile to exceed their contractually permitted credit limits, the interface locks the record. A Sales Manager or Administrator profile must authenticating-session credentials to input a bypass override.
- **Point of Sale (POS Terminal):** Operational cashiers utilize a high-speed retail transaction interface layout to process immediate cash sales transactions, select active items, append standard 18% VAT rates automatically, and print physical till receipts.

B. Customer Returns & RMA Workflows

When a client requests an inventory return tracking pipeline, use the following sequence:

1. Go to **Sales → Customer Returns → Log Return**.
2. The system triggers a unique tracking identity tag formatted as **RMA-XXX**.
3. **Inspection Logging:** Specify the explicit return root-cause reason, material grading condition (Restockable vs. Damaged), and the physical disposition target (Discard pile, Repair laboratory, or Restock back to Warehouse).
4. **Restocking Charges:** Input custom restocking penalty fees if necessary. The core database computes customer credit note amounts automatically using the script:

$$\text{Refund Amount} = \text{Returned Items Value} - \text{Restocking Fee}$$

Upon formal validation approval, restockable items are incremented back into warehouse stock records, and a corresponding financial Credit Note is posted to the bookkeeping ledger.

8. Logistics, Export Shipments & Gate Passes

Coordinates global logistics networks, shipping freight metadata, and physical security parameters.

A. Export Logistics Manifests

For cross-border ocean or air container orders, go to [Logistics](#) → [Export Logistics](#) → [Track Shipment](#) :

- **Logistics Meta-Data Entry:** Capture the manifest Vessel Name, Voyage Code, Bill of Lading (B/L) index code, Port of Loading, Destination Port of Discharge, ETD, and ETA markers.
- **Carrier Tracking API Link:** Store the logistics carrier web tracking link inside the document framework for click-to-track container visibility.
- **CIF Financial Totals:** The calculator script aggregates distinct valuation strings to compute finalized Cost, Insurance, and Freight (CIF) values:

$$\text{CIF Value} = \text{FOB Product Value} + \text{Freight Cost} + \text{Insurance Premium}$$

B. Security Gate Pass Management

To prevent inventory shrinkage and cargo theft, no commercial transport vehicle can exit the factory perimeter without a signed digital Gate Pass profile:

1. Navigate to [Logistics](#) → [Gate Pass Management](#) → [Generate Gate Pass](#) .
2. Link the pass record directly to the source Invoice file or Warehouse Transfer Order sheet.
3. Log transport trailer license numbers, driver legal name, assisting helper names, shipping container seal codes, and an itemized manifest item breakdown list.
4. The structural physical security officer stationed at the perimeter checking point visually audits the truck carriage against the live electronic Gate Pass form before executing a status confirmation of **Checked Out**.

9. Finance, Bank Accounts & Petty Cash Ledger

Tracks corporate accounting movements, liquidity status, and daily income records.

- **Petty Cash Ledger:** Tracks small internal daily expenditure cycles across factory operations (e.g., ad-hoc firewood purchases, site maintenance refreshments, urgent tool repairs). All cash drawing items require mapping to cost categories and must be verified by a Manager profile.
- **Bank Reconciliation & Cheque Tracking:** Synchronizes inbound automated electronic wire transfers directly against outstanding customer receivable books. The Cheque Ledger tracks physical bank check statuses by capturing check reference serials, originating banks, maturity dates, and live status markers (*Pending, Cleared, or Bounced*).

- **Real-Time Daily Profit & Loss (P&L):** Aggregates invoiced sales values (gross revenue items) against active manufacturing batch processing records (BOM costs), vendor supplier bills, petty cash spending data, and active shift labor costs.

10. HR, Roster Shifts & Payroll Engine

Coordinates workforce directories, time logs, attendance parameters, and regulatory corporate payroll execution modules.

- **Attendance Synchronization:** Roster configurations establish organizational working parameters (e.g., *Day Shift*, *Night Shift*, *Weekend Cover* templates). The tracking database automatically syncs against physical employee biometric barcode/RFID card swipe events to log roster timing variances.
- **Leave Self-Service:** Workers submit leave requests directly via their user login panels. HR Managers review remaining balance matrices to click Approve/Reject tags.
- **Payroll Engine Calculations:** Automatically processes monthly compensation metrics using a standard calculation rule script:

$$\text{Net Pay} = (\text{Base Salary} + \text{OT Hours} \times \text{OT Rate} + \text{Bonuses}) - (\text{Deductions} + \text{EPF/ETF Employee Share})$$

Validated payslips are compiled as secure digital PDFs and pushed out to individual employee profiles for instantaneous download.

11. System Administration (Audit Logs, Imports & Settings)

Provides system infrastructure utilities for IT system engineers to track platform operational integrity and configuration profiles.

- **Bulk Data Import Engine:** Accessible via `System Admin → Data Import`. System developers can quickly drop formatted CSV or Excel template sheets to perform bulk database indexing tasks covering catalog product listings, supplier directories, historical client tracking cards, and initial opening inventory ledger lines.
- **Taxation & Notification Settings:** Used to define standard tax parameters (such as the default 18% VAT rule). It manages downstream SMS notification API configurations, which send automated shipping updates to export clients or push monthly release notifications to staff profiles upon payslip compilation.
- **Database Backups:** Allows administrative engineering operators to trigger manual schema snapshots or configure cron-job schedules for secure cloud-server data backups.